Rubber_band_analysis

Analysis of Field Data and AI-AI in Rubber-band dynamics

PENPAL Project

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1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

```
## Field data dimensions: 820 14
```

```
## Simulated data dimensions: 30 12
```

```
##
## Field data columns:
```

```
##  [1] "timestamp"            "user"                 "ai"
##  [4] "conversation_id"      "respondent_id"        "interaction_count"
##  [7] "llm_type"             "turn"                 "starter"
## [10] "user_embedding"       "ai_embedding"         "pct_turn"
## [13] "user_sentiment_score" "ai_sentiment_score"
```

```
##
## Simulated data columns:
```

```
##  [1] "turn"                 "user"                 "ai"
##  [4] "client_id"            "workshop_id"          "conversation_id"
##  [7] "timestamp"            "user_embedding"       "ai_embedding"
## [10] "pct_turn"             "user_sentiment_score" "ai_sentiment_score"
```

1.2 1.2 Data Preprocessing

```
##
## === Field Data Summary ===
```

```
## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1              81     729                    9            0.0426           0.171
```

```
##
## === Simulated Data Summary ===
```

```
## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1               6      30                    5            -0.407          -0.296
```

1.3 1.3 Compute Lagged Variables and Cosine

```
## === Lag Variable Verification ===
```

```
## Field data - Turn 1 with non-NA lag (should be 0): 0
```

```
## Simulated data - Turn 1 with non-NA lag (should be 0): 0
```

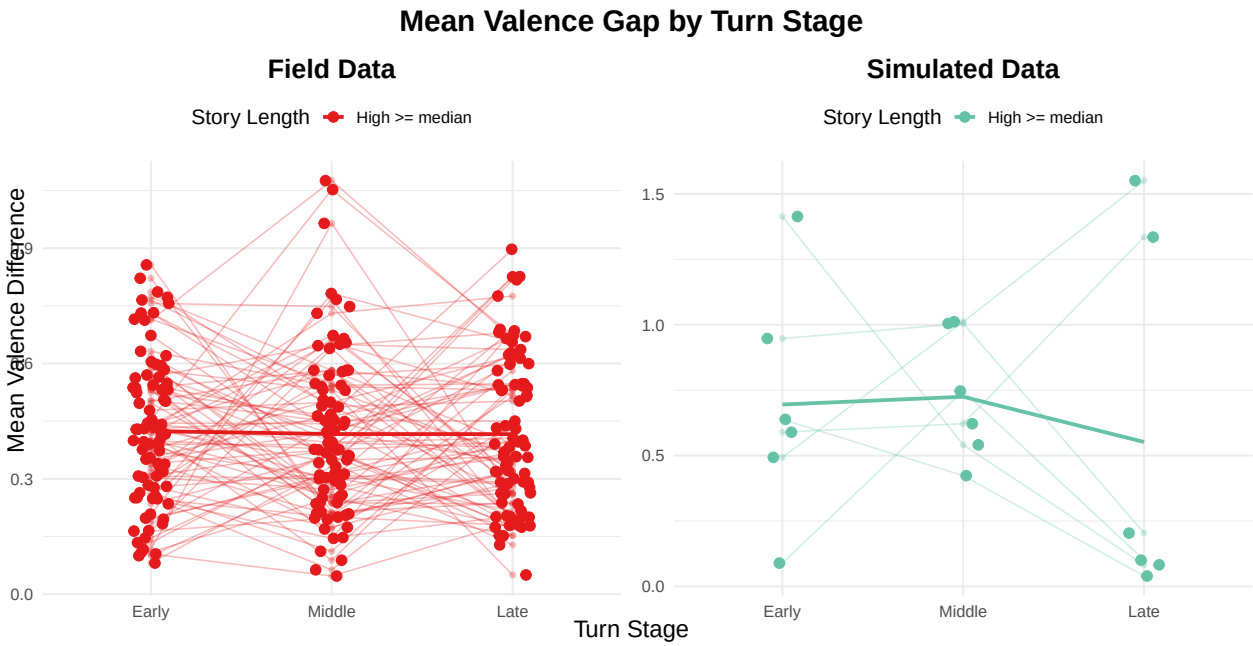
2 2. Temporal Dynamics Analysis

2.1 2.1 Stage-Based Analysis (Early/Middle/Late)

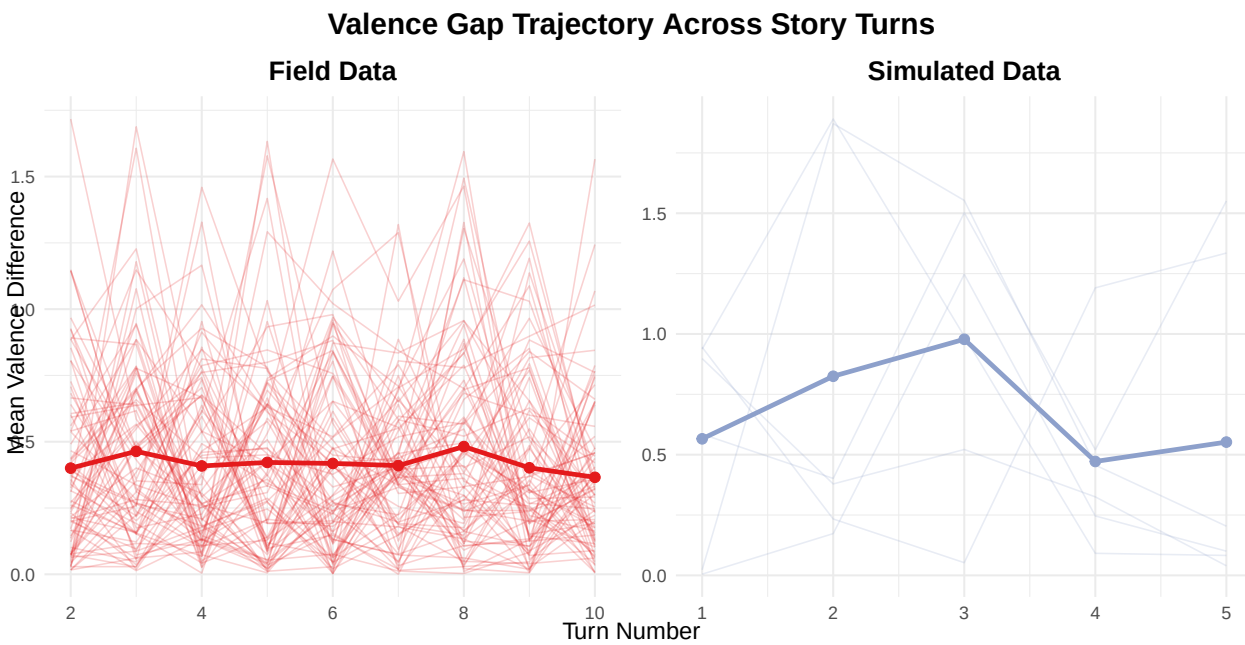
```
## === Mean Valence Gap by Stage ===
```

```
## # A tibble: 6 x 4
##   data_source stage_label mean_gap sd_gap
##   <chr>       <fct>          <dbl>  <dbl>
## 1 Field       Early          0.424     NA
## 2 Field       Middle         0.417     NA
## 3 Field       Late           0.416     NA
## 4 Simulated   Early          0.695     NA
## 5 Simulated   Middle         0.725     NA
## 6 Simulated   Late           0.552     NA
```

2.2 2.2 Stage Valence Gap Plot



2.3 2.3 Turn-by-Turn Valence Gap Trajectory



2.4 2.4 Rubber-Band Dynamics

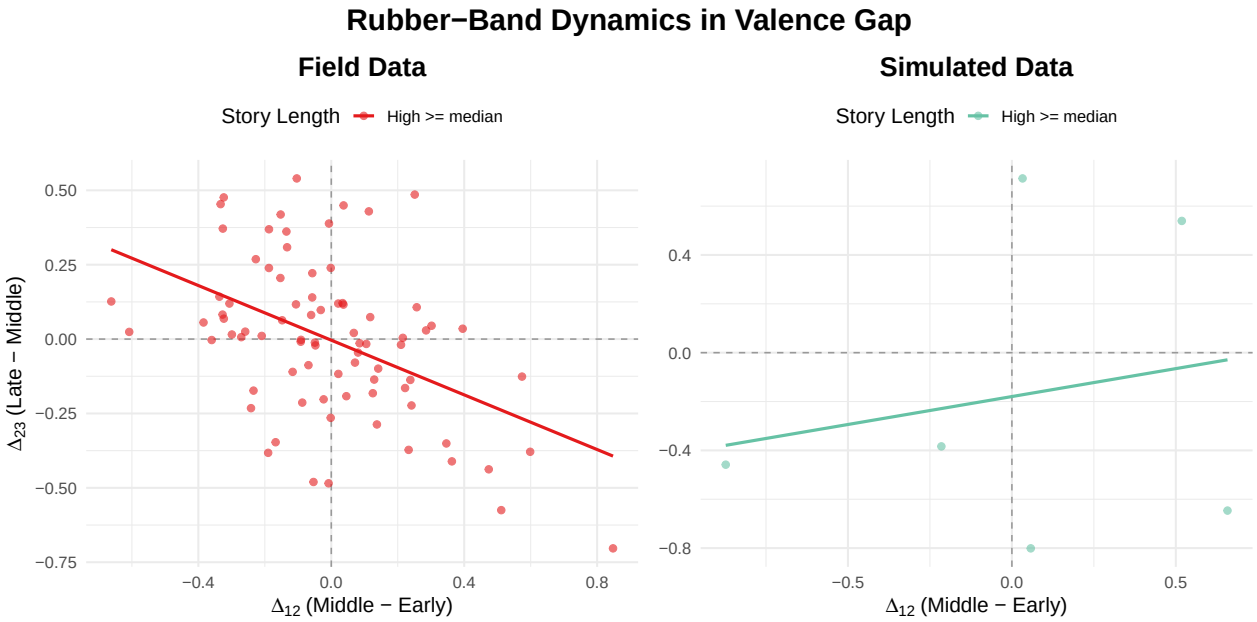
```
## === Rubber-Band Model: Field Data ===

##
## Call:
## lm(formula = delta_23 ~ delta_12, data = rb_field)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.50112 -0.15980 -0.03059  0.14131  0.60477
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.003751   0.026065  -0.144    0.886
## delta_12    -0.459356   0.098798  -4.649 1.31e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2345 on 79 degrees of freedom
## Multiple R-squared:  0.2148, Adjusted R-squared:  0.2049
## F-statistic: 21.62 on 1 and 79 DF,  p-value: 1.314e-05

##
## === Rubber-Band Model: Simulated Data ===
```

2.5 2.5 Turn-by-Turn Rubber-Band Dynamics

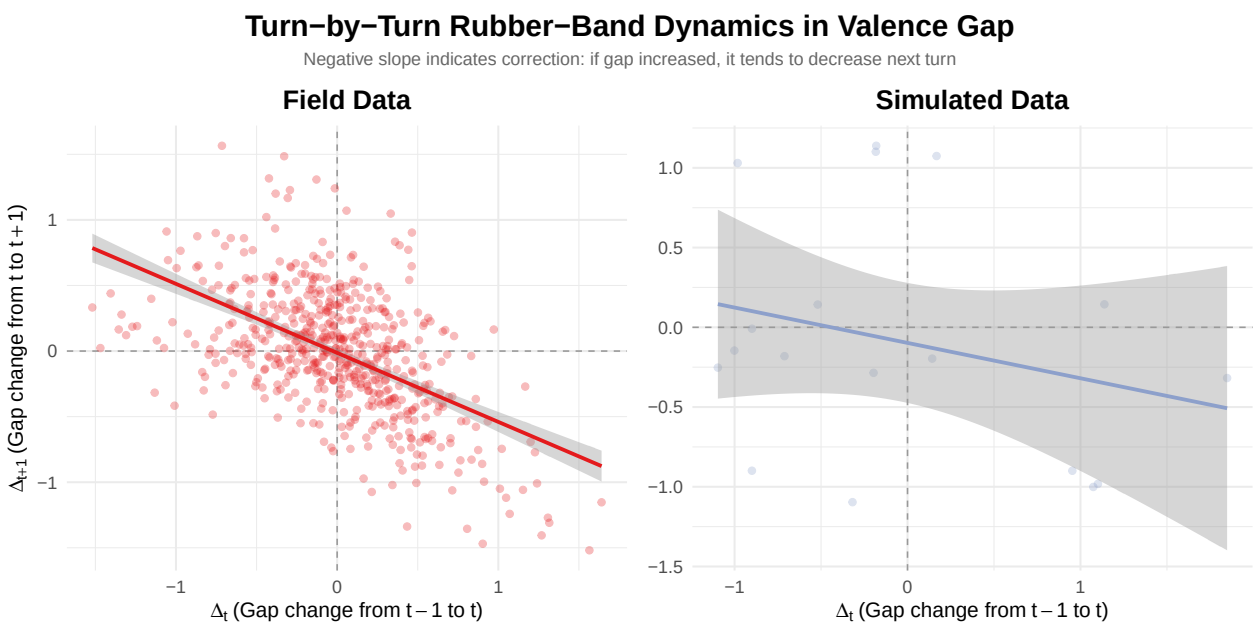
```
## === Turn-Level Rubber-Band Model: Field Data ===

##
## Call:
## lm(formula = gap_change_next ~ gap_change, data = rb_turns_field)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.09551 -0.26469 -0.02615  0.25963  1.32611
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.01389    0.01676  -0.829    0.407
## gap_change  -0.52623    0.03509 -14.995   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.399 on 565 degrees of freedom
## Multiple R-squared:  0.2847, Adjusted R-squared:  0.2834
## F-statistic: 224.8 on 1 and 565 DF,  p-value: < 2.2e-16

##
## === Turn-Level Rubber-Band Model: Simulated Data ===

##
## Call:
## lm(formula = gap_change_next ~ gap_change, data = rb_turns_sim)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.0692 -0.5424 -0.1708  0.4181  1.2092
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.09783    0.17752  -0.551    0.589
## gap_change  -0.22171    0.20273  -1.094    0.290
##
## Residual standard error: 0.7527 on 16 degrees of freedom
## Multiple R-squared:  0.06955,    Adjusted R-squared:  0.0114
## F-statistic: 1.196 on 1 and 16 DF,  p-value: 0.2903

##
## === Turn-Level Rubber-Band Correlation ===

## Field correlation (gap_change vs gap_change_next): -0.5335425

## Simulated correlation (gap_change vs gap_change_next): -0.2637313
```

2.6 2.6 Rubber-Band Dynamics Plot



2.7 2.7 Turn-Level Rubber-Band Dynamics Plot



3 3. Permutation tests

3.1 3.1 Stage-level Permutation Test

```
## === Stage-Level Permutation Test Results ===
## Observed slope: -0.4594
## Null mean: -0.5001
## Null SD: 0.0988
```

```
## Effect size (z): 0.41
## P-value: 0.649
```

3.2 3.2 Brutal Stage-Level Permutation

```
## === Stage-Level Permutation Test Results - Shuffle Turn-Level Valence Gap first ===
## Observed slope: -0.4594
## Null mean: -0.5022
## Null SD: 0.1024
## Effect size (z): 0.42
## P-value: 0.67
```

3.3 3.2 Global Shuffle Stage-Level Test

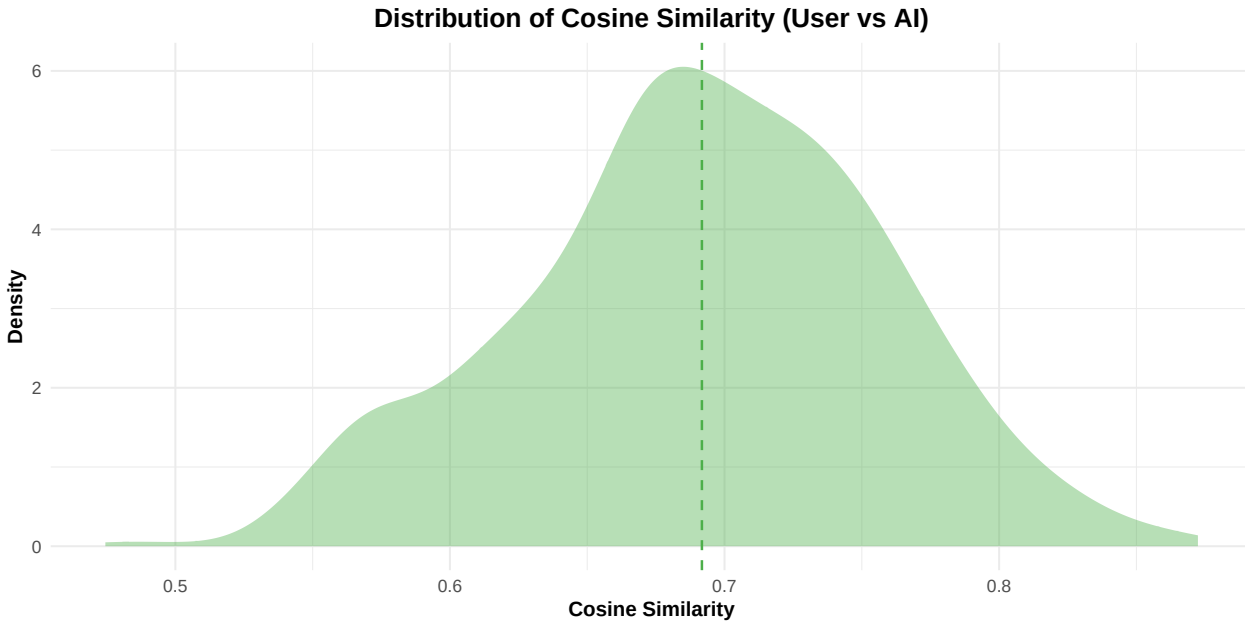
```
##
## Call:
## lm(formula = d23 ~ d12, data = df_global)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.58127 -0.17615 -0.00051  0.16275  0.51375
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.06627    0.05872   1.129   0.2731
## d12         -0.64874    0.33162  -1.956   0.0653 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2667 on 19 degrees of freedom
## Multiple R-squared:  0.1677, Adjusted R-squared:  0.1238
## F-statistic: 3.827 on 1 and 19 DF,  p-value: 0.0653
```

3.4 3.3 Turn-Level Permutation Test

```
## === Turn-Level Permutation Test Results ===
## Observed slope: -0.5262
## Null mean: -0.0025
## Null SD: 0.0577
## Effect size (z): -9.08
## P-value: <2e-16
## # A tibble: 2 x 4
##   transition_type rb_slope correlation     n
##   <chr>              <dbl>       <dbl> <int>
## 1 Stage Boundary    -0.516      -0.501   162
## 2 Within-Stage      -0.564      -0.579   324
```

4 4. Cosine similarity alignment

4.1 4.1 Cosine Similarity Distribution

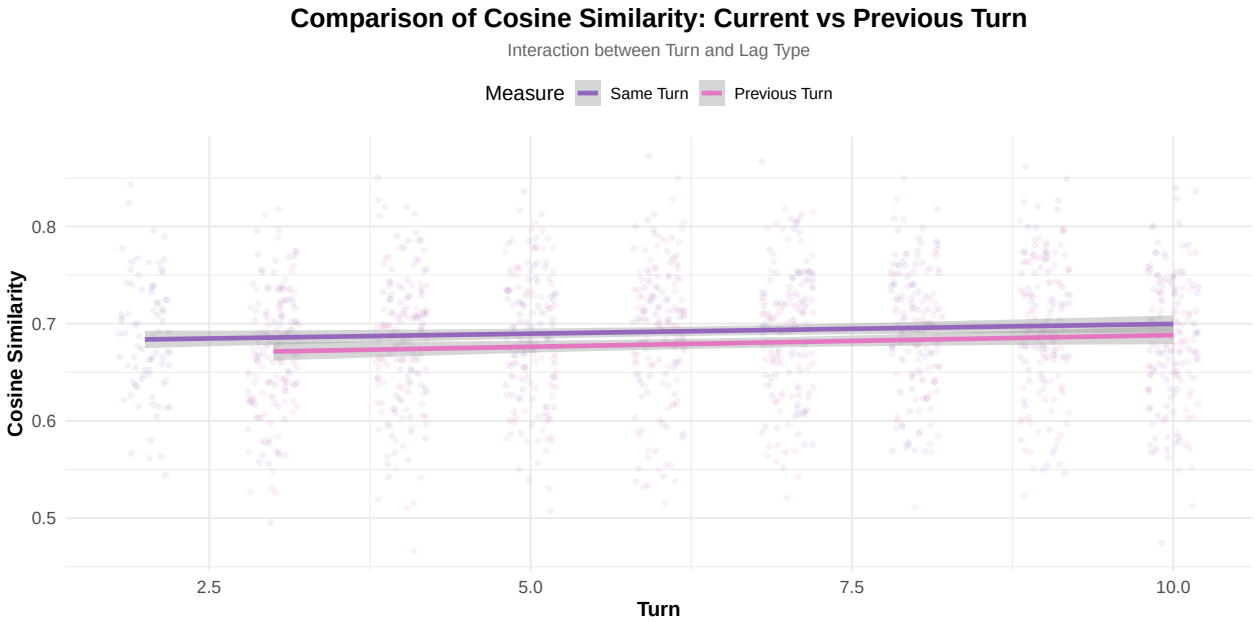


4.2 4.2 Cosine Similarity Mixed Model (Interaction)

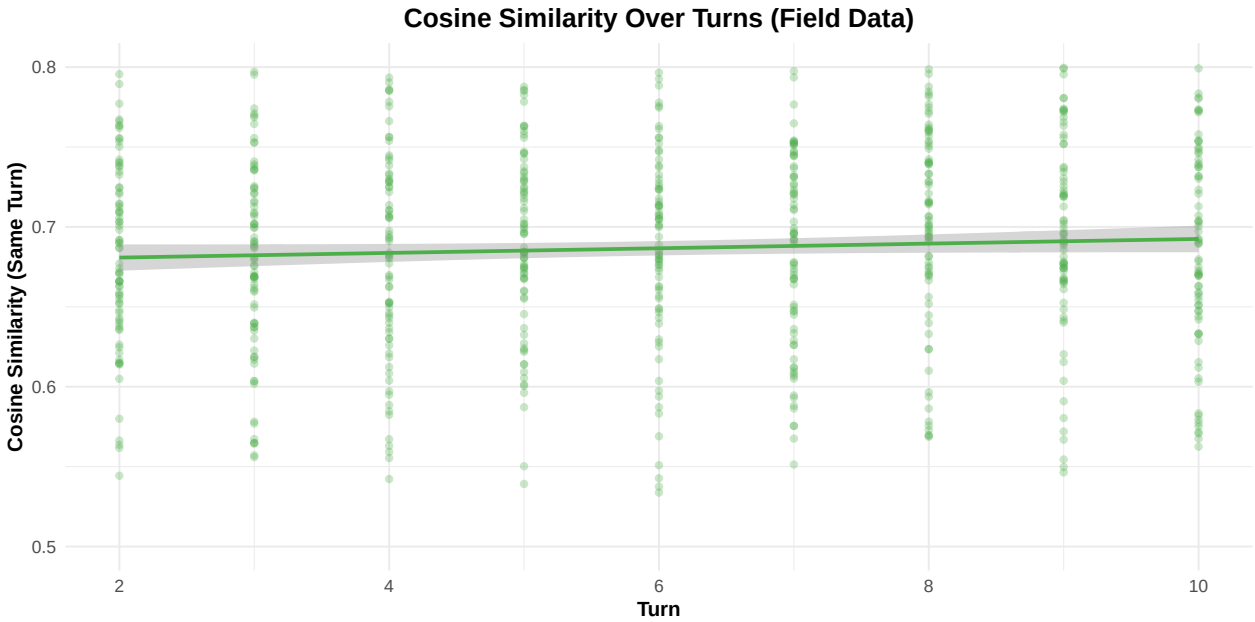
```
## === Cosine Similarity Interaction Model (Same vs Previous Turn) ===

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: cosine_sim ~ turn * lag_type + (1 | conversation_id)
##    Data: df_cosine_long
##
## REML criterion at convergence: -3665.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.5471 -0.6805  0.0229  0.6839  3.0047
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0007713 0.02777
##  Residual                    0.0036315 0.06026
## Number of obs: 1377, groups:  conversation_id, 81
##
## Fixed effects:
##                              Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)                 6.798e-01  6.435e-03  6.831e+02 105.646   <2e-16
## turn                        1.996e-03  8.644e-04  1.293e+03   2.309   0.0211
## lag_typePrevious Turn      -1.558e-02  9.088e-03  1.293e+03  -1.714   0.0867
## turn:lag_typePrevious Turn  4.099e-04  1.347e-03  1.293e+03   0.304   0.7609
##
## (Intercept)                ***
## turn                       *
```

```
## lag_typePrevious Turn      .
## turn:lag_typePrevious Turn
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) turn   lg_tPT
## turn        -0.806
## lg_typPrvsT -0.545  0.571
## trn:lg_tyPT  0.517 -0.642 -0.933
```

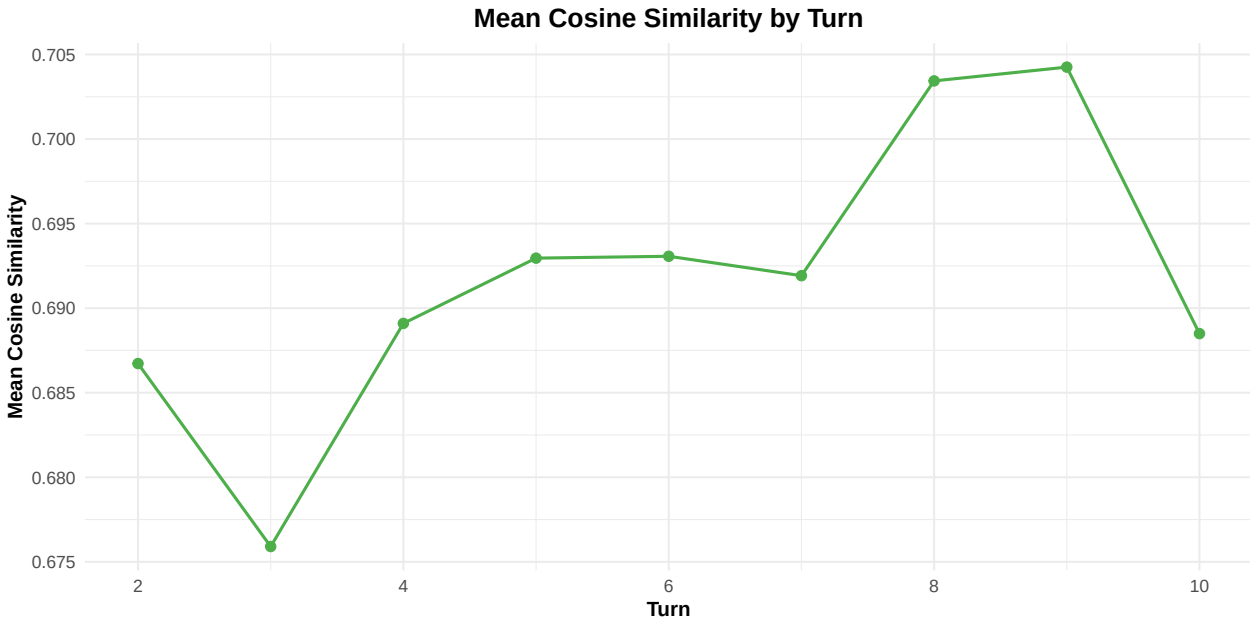
4.3 4.2b Comparison of Current vs Previous Turn Cosine Similarity

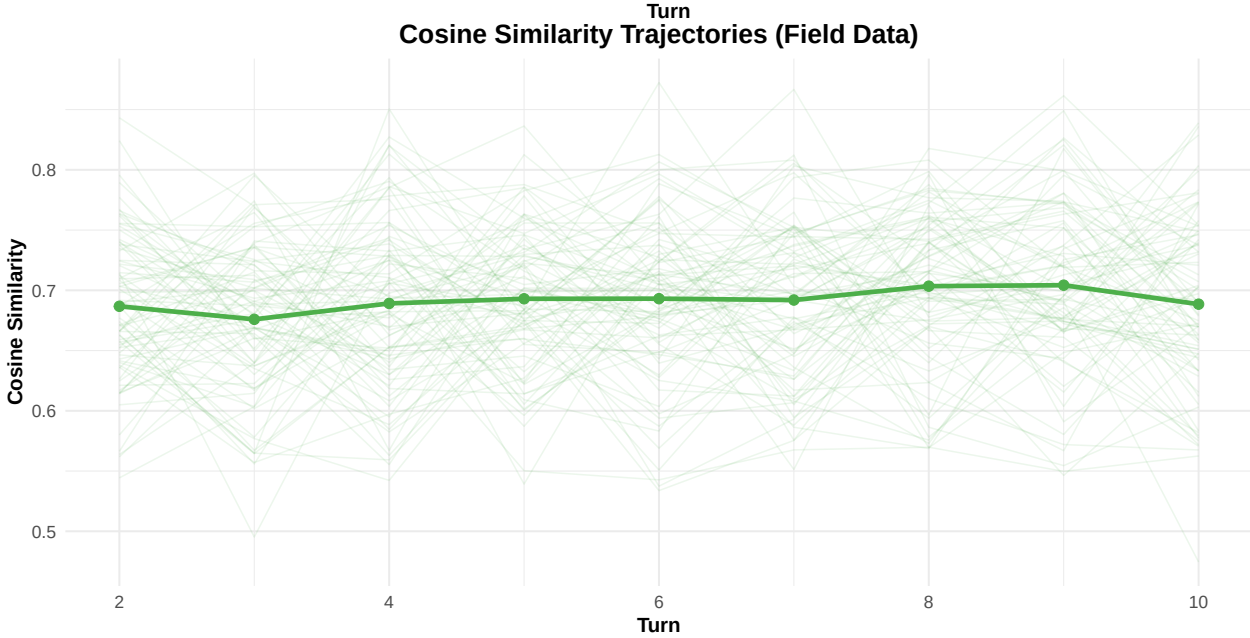


4.4 4.3 Cosine Similarity Linear Plot



4.5 4.4 Cosine Similarity Over Time





4.6 4.5 Cosine rubber band

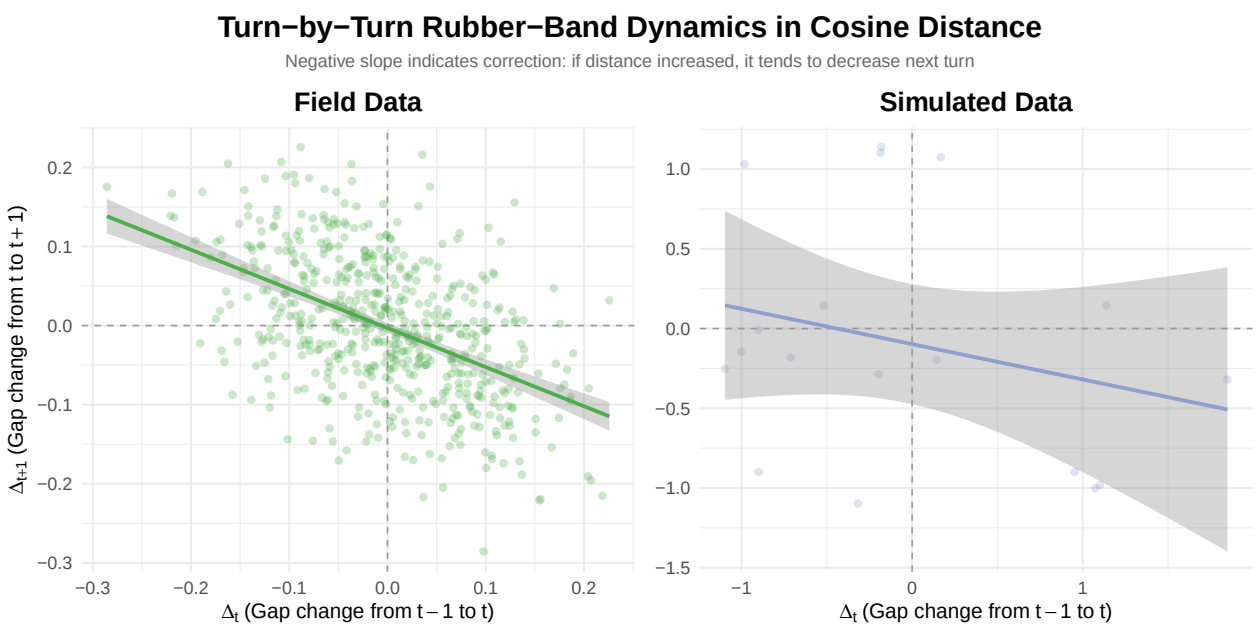
```
## === Turn-Level Rubber-Band Cosine Distance: Field Data ===

##
## Call:
## lm(formula = dist_change_next ~ dist_change, data = rb_cosine)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.234210 -0.054262  0.000174  0.053382  0.236784
##
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.003038   0.003217  -0.944    0.345
## dist_change -0.494722   0.037627 -13.148   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.07657 on 565 degrees of freedom
## Multiple R-squared:  0.2343, Adjusted R-squared:  0.2329
## F-statistic: 172.9 on 1 and 565 DF,  p-value: < 2.2e-16
```

4.7 4.6 Turn level Cosine Permutation test

```
## === Cosine Distance Turn-Level Permutation Test Results ===

## Observed slope: -0.4947

## Null mean: -3e-04

## Null SD: 0.0554

## Effect size (z): -8.93

## P-value: <2e-16

## # A tibble: 2 x 4
##   transition_type rb_slope correlation     n
##   <chr>              <dbl>       <dbl> <int>
## 1 Stage Boundary    -0.500      -0.507   162
## 2 Within-Stage      -0.497      -0.502   324
```

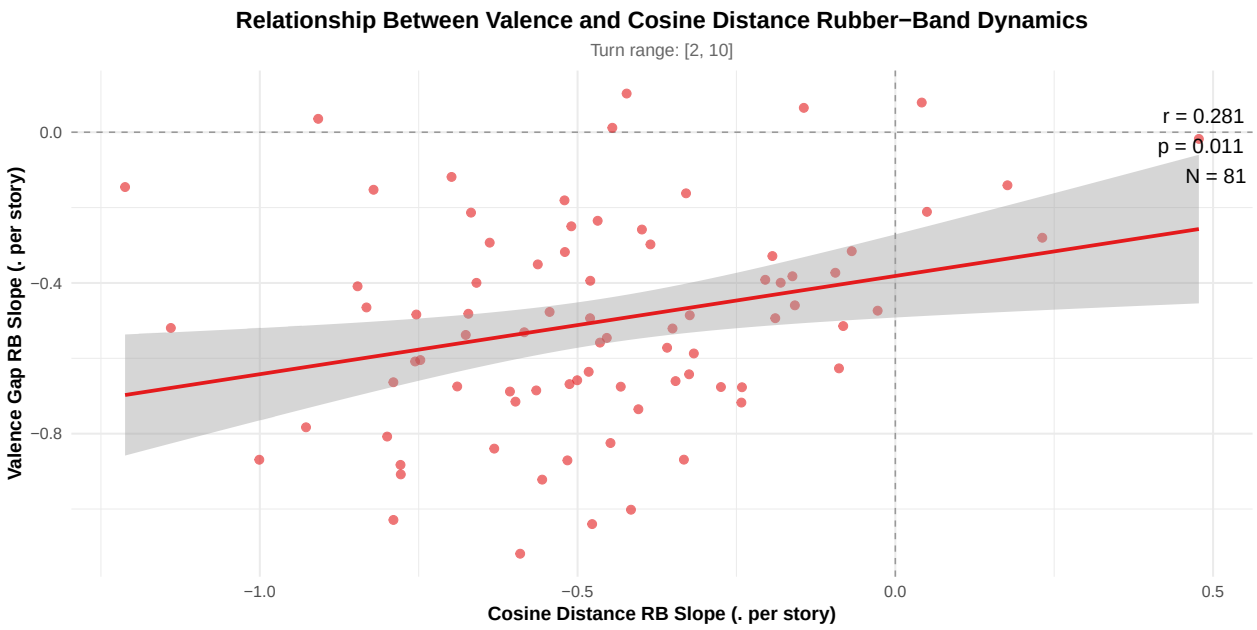
4.8 4.7 Turn Level Cosine RB Linear plot



4.9 4.8 Correlation Between Cosine Distance and Valence Rubber-Band

```
## === Correlation Between Valence and Cosine Distance Rubber-Band ===

## Turn range: [2, 10]
```

```
## N observations (turn-level): 567
## Turn-level correlation of gap changes (Δt): 0.032
##   Interpretation: Do valence and cosine gaps change in the same direction
##                   at the same turn across all stories?
## Turn-level correlation of next changes (Δt+1): 0.024
##   Interpretation: Do subsequent valence and cosine corrections correlate?
## N stories with >= 3 turns: 81
## N stories with >= 3 turns: 81
## STORY-LEVEL correlation of RB slopes: 0.281
##   Interpretation: Across stories, does valence RB strength predict cosine RB strength?
##                   Positive r means stories with strong valence correction
##                   also show strong cosine distance correction.
##   95% CI: [0.067, 0.47]
##   p-value: 0.0111
##   df: 79
## Shared variance (R²): 0.079
## Interpretation: 7.9 % of variation in valence RB slope
##                 is explained by cosine distance RB slope
```




```
##
## === Regression: Cosine RB Predicting Valence RB ===
##
## Call:
## lm(formula = valence_rb_slope ~ cosine_rb_slope, data = rb_slopes)
##
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
## -0.58320 -0.17774 -0.02833  0.17737  0.65372
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     -0.38151    0.05539  -6.887 1.23e-09 ***
## cosine_rb_slope  0.26064    0.10020   2.601   0.0111 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2706 on 79 degrees of freedom
## Multiple R-squared:  0.07889,    Adjusted R-squared:  0.06723
## F-statistic: 6.766 on 1 and 79 DF,  p-value: 0.01109

##
## === VALIDATION CHECK ===

## Correlation from cor(): 0.281

## Correlation from lm R²: 0.281

## These should match (they do if correlation is positive).
```

5 5. Supplementary: Interaction Effects

5.1 5.1 Starter Interaction Effects

5.1.1 5.1.1 Valence Gap by Starter

```
## === Valence Rubber-Band: Starter Interaction ===

##
## Call:
## lm(formula = gap_change_next ~ gap_change * as.factor(starter),
##     data = rb_turns_field)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.10611 -0.26434 -0.02794  0.25699  1.33266
##
## Coefficients:
##                                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      -0.013053   0.023594  -0.553    0.580
## gap_change                       -0.503753   0.050678  -9.940   <2e-16 ***
## as.factor(starter)user           -0.001125   0.033575  -0.034    0.973
## gap_change:as.factor(starter)user -0.043266   0.070365  -0.615    0.539
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3996 on 563 degrees of freedom
## Multiple R-squared:  0.2851, Adjusted R-squared:  0.2813
## F-statistic: 74.86 on 3 and 563 DF,  p-value: < 2.2e-16
```

5.1.2 5.1.2 Cosine Distance by Starter

```
## === Cosine Distance Rubber-Band: Starter Interaction ===
```

```
##
## Call:
## lm(formula = dist_change_next ~ dist_change * starter, data = rb_cosine)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.242853 -0.052551 -0.001561  0.055063  0.232948
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)             -0.004126   0.004516  -0.914   0.3613
## dist_change             -0.559973   0.051013 -10.977   <2e-16 ***
## starteruser              0.002174   0.006425   0.338   0.7353
## dist_change:starteruser  0.142508   0.075418   1.890   0.0593 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.07646 on 563 degrees of freedom
## Multiple R-squared:  0.2392, Adjusted R-squared:  0.2352
## F-statistic: 59.01 on 3 and 563 DF,  p-value: < 2.2e-16
```

5.1.3 5.1.3 Alignment (Cosine Similarity) by Starter

```
## === Cosine Similarity (Alignment): Starter × Turn Interaction ===
```

```
##
## Call:
## lm(formula = cosine_sim_same ~ turn * starter, data = df_alignment_starter)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.225898 -0.040856  0.001462  0.046601  0.182698
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)       0.6756066  0.0087110  77.558   <2e-16 ***
## turn              0.0023498  0.0013336   1.762   0.0785 .
## starteruser       0.0084614  0.0123959   0.683   0.4951
## turn:starteruser -0.0007162  0.0018977  -0.377   0.7060
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06614 on 725 degrees of freedom
## Multiple R-squared:  0.007245,   Adjusted R-squared:  0.003137
## F-statistic: 1.764 on 3 and 725 DF,  p-value: 0.1527
```

5.2 5.2 LLM Type Interaction Effects

5.2.1 5.2.1 Valence Gap by LLM Type

```
## === Valence Rubber-Band: LLM Type Interaction ===
```

```
##
## Call:
## lm(formula = gap_change_next ~ gap_change * factor(llm_type),
##     data = rb_turns_llm)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.04664 -0.25770 -0.01034  0.24751  1.28309
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   0.02551    0.03562   0.716    0.474
## gap_change                   -0.53725    0.07244  -7.416 4.49e-13 ***
## factor(llm_type)2            -0.03485    0.05038  -0.692    0.489
## factor(llm_type)3            -0.05069    0.04712  -1.076    0.282
## factor(llm_type)4            -0.06302    0.04854  -1.298    0.195
## gap_change:factor(llm_type)2  0.02949    0.10705   0.275    0.783
## gap_change:factor(llm_type)3  0.08205    0.10159   0.808    0.420
## gap_change:factor(llm_type)4 -0.04716    0.09537  -0.494    0.621
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3997 on 559 degrees of freedom
## Multiple R-squared:  0.2896, Adjusted R-squared:  0.2807
## F-statistic: 32.55 on 7 and 559 DF,  p-value: < 2.2e-16
```

5.2.2 5.2.2 Cosine Distance by LLM Type

```
## === Cosine Distance Rubber-Band: LLM Type Interaction ===
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## dist_change_next ~ dist_change * factor(llm_type) + (1 | conversation_id)
##    Data: rb_cosine_llm
##
## REML criterion at convergence: -1262.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.00982 -0.68353 -0.01916  0.72215  3.12761
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.000000 0.00000
##  Residual                    0.005905 0.07685
## Number of obs: 567, groups:  conversation_id, 81
##
## Fixed effects:
##                                 Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)                   -1.796e-03  6.847e-03  5.590e+02  -0.262    0.793
## dist_change                   -5.373e-01  7.189e-02  5.590e+02  -7.474 3.02e-13
## factor(llm_type)2             -8.626e-04  9.683e-03  5.590e+02  -0.089    0.929
## factor(llm_type)3             -3.640e-03  9.066e-03  5.590e+02  -0.401    0.688
## factor(llm_type)4             -1.667e-04  9.331e-03  5.590e+02  -0.018    0.986
## dist_change:factor(llm_type)2  1.310e-01  1.096e-01  5.590e+02   1.195    0.233
## dist_change:factor(llm_type)3  9.975e-03  9.869e-02  5.590e+02   0.101    0.920
## dist_change:factor(llm_type)4  5.774e-02  1.104e-01  5.590e+02   0.523    0.601
##
```

```
## (Intercept)
## dist_change                   ***
## factor(llm_type)2
## factor(llm_type)3
## factor(llm_type)4
## dist_change:factor(llm_type)2
## dist_change:factor(llm_type)3
## dist_change:factor(llm_type)4
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) dst_ch fc(_)2 fc(_)3 fc(_)4 d_:(_)2 d_:(_)3
## dist_change  0.021
## fctr(llm_)2 -0.707 -0.015
## fctr(llm_)3 -0.755 -0.016  0.534
## fctr(llm_)4 -0.734 -0.015  0.519  0.554
## dst_ch:(_)2 -0.014 -0.656  0.009  0.010  0.010
## dst_ch:(_)3 -0.015 -0.728  0.011  0.041  0.011  0.478
## dst_ch:(_)4 -0.014 -0.651  0.010  0.010  0.020  0.427   0.474
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

5.2.3 5.2.3 Alignment (Cosine Similarity) by LLM Type

```
## === Cosine Similarity (Alignment): LLM Type × Turn Interaction ===
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: cosine_sim_same ~ turn * factor(llm_type) + (1 | conversation_id)
##    Data: df_alignment_llm
##
## REML criterion at convergence: -1858.8
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.6932 -0.6487  0.0068  0.6837  2.7134
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0005218 0.02284
##  Residual                    0.0038110 0.06173
## Number of obs: 729, groups:  conversation_id, 81
##
## Fixed effects:
##                          Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)             6.720e-01  1.340e-02  5.296e+02  50.152   <2e-16 ***
## turn                    2.075e-03  1.878e-03  6.440e+02   1.105    0.270
## factor(llm_type)2       2.067e-02  1.895e-02  5.296e+02   1.091    0.276
## factor(llm_type)3       1.181e-02  1.773e-02  5.296e+02   0.666    0.506
## factor(llm_type)4      -1.206e-03  1.826e-02  5.296e+02  -0.066    0.947
## turn:factor(llm_type)2 -7.742e-04  2.657e-03  6.440e+02  -0.291    0.771
## turn:factor(llm_type)3  7.692e-04  2.485e-03  6.440e+02   0.310    0.757
## turn:factor(llm_type)4 -5.211e-04  2.560e-03  6.440e+02  -0.204    0.839
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) turn   fc(_)2 fc(_)3 fc(_)4 t:(_)2 t:(_)3
## turn        -0.841
## fctr(llm_)2 -0.707  0.595
## fctr(llm_)3 -0.756  0.636  0.535
## fctr(llm_)4 -0.734  0.617  0.519  0.555
## trn:fct(_)2  0.595 -0.707 -0.841 -0.450 -0.436
## trn:fct(_)3  0.636 -0.756 -0.450 -0.841 -0.467  0.535
## trn:fct(_)4  0.617 -0.734 -0.436 -0.467 -0.841  0.519  0.555
```